

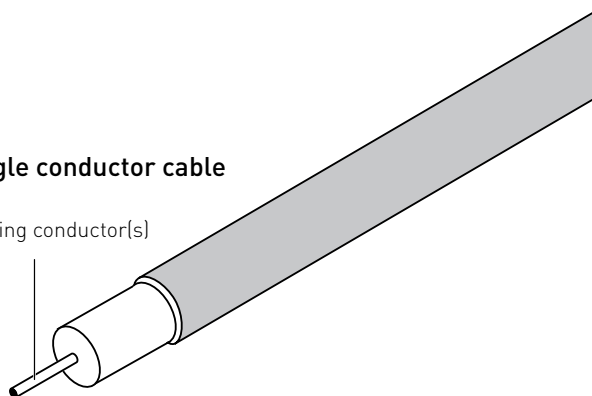
Raychem HAx

MINERAL INSULATED (MI) ALLOY 825 HEATING CABLE

TYPICAL CABLE CONSTRUCTIONS

Single conductor cable

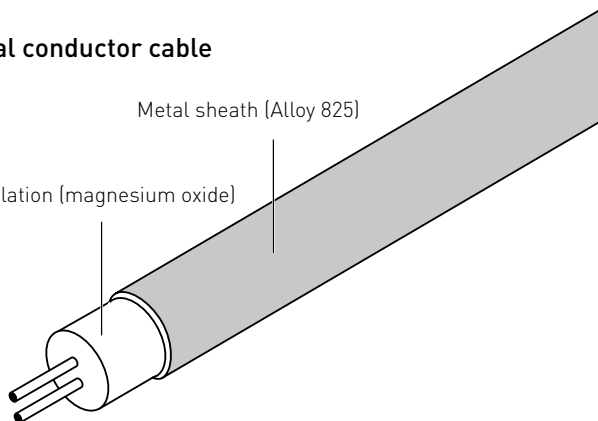
Heating conductor[s]



Dual conductor cable

Metal sheath (Alloy 825)

Insulation (magnesium oxide)



Raychem HAx mineral insulated (MI) Alloy 825 series heating cables are suitable for use in hazardous areas. They have been designed for use in freeze protection and temperature maintenance applications of pipes, tanks and other equipment.

MI heating cables of the HAx-series offer an ideal combination of ruggedness, high temperature withstand capability and corrosion resistance and can therefore be used for a wide variety of heat-tracing applications, in particular for applications with high power requirements and for temperatures exceeding the capabilities of polymer insulated (PI) series heating cables.

The heating cables can be used for exposure temperatures of up to 700°C and a typical power output of up to 270 W/m. Higher temperatures and power outputs can be achieved, contact Pentair for assistance.

HAx mineral insulated (MI) heating cables are available as single and dual conductor construction and in a very wide range of resistances. The use of dual conductor heating cables can significantly reduce total installed cost and simplifies installation, in particular for small pipes and instrument tubing.

The heating cables are offered as bulk cable as well as factory terminated heating units employing brazing and laser welding technology. The offering is completed with a full range of components for installation, connection and splicing of the heating cables.

APPLICATION

Area classification	Hazardous area, Zone 1 or Zone 2 (Gas) or Zone 21 or zone 22 (Dust) Ordinary
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APPROVALS

System (heating units)

Baseefa02ATEX0046X

 II 2GD Ex e II T6 to T1 Ex tD A21 IP6X

EARC

Ex

TC RU C-BE.MI062.B.02787

1 Ex e II T6...T1 Gb X

Ex td A21 IP66 T6...T1

Bulk cable

Baseefa02ATEX0045U

 II 2G Ex e II

EARC

Ex

TC RU C-BE.ME92.B.00056

1 Ex e II T6...T1 Gb X

Ex td A21 IP66 T6...T1

TECHNICAL DATA

Cable sheath material	Alloy 825		
Conductor material	Various alloys and copper		
Max. exposure temperature	550°C (brazed heating units) 700°C* (laser welded heating units) *Higher temperatures can be realized, contact Pentair		
Min. installation temperature	-60°C		
Min. bending radius	6 x OD (cable diameter) at -60°C		
Max. supply voltage and power	Voltage (U_0/U)	Max. power output*	Heating cable type
	600/600 Vac	210 W/m	HAX1N Single conductor cable, 600 V
	300/300 Vac	200 W/m	HAX2M Dual conductor cable, 300 V
	600/600 Vac	270 W/m	HAX2N Dual conductor cable, 600 V
*typical value, depending on application			
Earth leakage	3 mA /100 m (nominal at 20°C, 230 Vac, 50 - 60 Hz)		
Min. cable spacing	25 mm for hazardous areas		

TABLE 1 MI SERIES HEATING CABLES HAX2M (Dual conductor cable, 300 V)

Order Reference	Nominal resistance (Ω/km @ 20°C)	Outer diameter (mm)	Temp. coefficient ($\times 10^{-3}/\text{K}$)	Max. coil length [m]	Nom. weight (kg/km)	Part Number PN
HAF2M59K	59000	4.4	0.09	387	73	32SF1180
HAF2M36K	36000	4.0	0.09	483	60	32SF1110
HAF2M29.5K	29500	4.1	0.09	459	63	32SF2900
HAF2M24.5K	24500	4.0	0.09	477	61	32SF2750
HAA2M19.7K	19700	4.1	0.09	459	63	32SA2600
HAA2M13.2K	13200	3.7	0.09	554	54	32SA2400
HAA2M10.4K	10400	4.4	0.09	389	74	32SA2318
HAA2M9000	9000	3.9	0.09	505	60	32SA2275
HAA2M6600	6600	4.3	0.09	414	73	32SA2200
HAA2M5600	5600	4.2	0.09	425	72	32SA2170
HAB2M3750	3750	4.4	0.04	390	76	32SB2114
HAB2M3000	3000	4.1	0.04	451	67	32SB3914
HAB2M2300	2300	4.3	0.04	411	74	32SB3700
HAQ2M1560	1560	4.5	0.5	376	78	32SQ3472
HAQ2M1240	1240	4.6	0.5	352	82	32SQ3374
HAQ2M965	965	4.5	0.5	368	79	32SQ3293
HAQ2M660	660	4.1	0.5	457	66	32SQ3200
HAQ2M495	495	4.3	0.5	420	73	32SQ3150
HAQ2M330	330	4.7	0.5	348	89	32SQ3100
HAP2M240	240	4.4	1.3	391	78	32SP4734
HAP2M190	190	4.5	1.3	375	82	32SP4583
HAP2M150	150	4.8	1.3	337	62	32SP4458
HAC2M105	105	4.7	3.9	349	85	32SC4324

Resistance tolerance: $\pm 10\%$ **TABLE 2 MI SERIES HEATING CABLES HAX2N** (Dual conductor cable, 600 V)

Order Reference	Nominal resistance (Ω/km @ 20°C)	Outer diameter (mm)	Temp. coefficient ($\times 10^{-3}/\text{K}$)	Max. coil length [m]	Nom. weight (kg/km)	Part Number PN
HAF2N36K	36000	4.9	0.09	312	91	62SF1110
HAF2N29.5K	29500	4.9	0.09	312	91	62SF2900
HAF2N24.5K	24500	5.2	0.09	279	103	62SF2750
HAF2N19.7K	19700	5.8	0.09	222	128	62SF2600
HAA2N13.6K	13600	6.1	0.09	204	140	62SA2414
HAA2N9000	9000	5.7	0.09	232	125	62SA2275
HAF2N6600	6600	6.2	0.09	196	149	62SF2200
HAA2N5600	5600	6.1	0.09	205	143	62SA2170
HAT2N3750	3750	5.5	0.18	254	113	62ST2115

TABLE 2 MI SERIES HEATING CABLES HAX2N (Dual conductor cable, 600 V)

Order Reference	Nominal resistance (Ω/km @ 20°C)	Outer diameter (mm)	Temp. coefficient ($\times 10^{-3}/\text{K}$)	Max. coil length [m]	Nom. weight (kg/km)	Part Number PN
HAB2N3000	3000	5.9	0.04	219	132	62SB3914
HAB2N2300	2300	6.7	0.04	168	174	62SB3700
HAT2N1670	1670	5.5	0.18	255	115	62ST3505
HAQ2N1240	1240	5.5	0.5	254	113	62SQ3374
HAQ2N940	940	5.6	0.5	239	121	62SQ3286
HAQ2N660	660	5.8	0.5	229	128	62SQ3200
HAQ2N495	495	5.8	0.5	229	128	62SQ3150
HAQ2N330	330	6.5	0.5	179	165	62SQ3100
HAP2N255	255	6.4	1.3	188	155	62SP4775
HAP2N185	185	6.7	1.3	171	173	62SP4561
HAP2N130	130	7.0	1.3	154	194	62SP4402
HAP2N92	92	7.4	1.3	139	219	62SP4281
HAC2N66	66	7.2	3.9	145	201	62SC4200
HAC2N43	43	7.7	3.9	128	233	62SC4130
HAC2N27	27	8.4	3.9	100	279	62SC5818
HAC2N17	17	9.2	3.9	90	343	62SC5516
HAC2N10.5	10.5	10.2	3.9	74	432	62SC5324
HAC2N6.6	6.6	12.6	3.9	48	653	62SC5204
HAC2N4.3	4.3	13.8	3.9	143	769	62SC5128

TABLE 3 MI SERIES HEATING CABLES HAX1N (Single conductor cable, 600 V)

Order Reference	Nominal resistance (Ω/km @ 20°C)	Outer diameter (mm)	Temp. coefficient ($\times 10^{-3}/\text{K}$)	Max. coil length [m]	Nom. weight (kg/km)	Part Number PN
HAA1N6565	6565	4.3	0.085	406	75	61SA2200
HAA1N5250	5250	4.1	0.085	443	66	61SA2160
HAA1N4300	4300	4.1	0.085	460	63	61SA2130
HAA1N3300	3300	4.1	0.085	460	64	61SA2100
HAA1N2800	2800	4.3	0.085	408	72	61SA3850
HAA1N2300	2300	4.1	0.085	462	64	61SA3700
HAA1N1640	1640	4.3	0.085	410	73	61SA3500
HAT1N920	920	4.3	0.18	408	72	61ST3280
HAB1N660	660	4.6	0.04	365	82	61SB3200
HAB1N500	500	4.3	0.04	412	76	61SB3150
HAQ1N390	390	4.4	0.5	384	75	61SQ3118
HAQ1N240	240	4.3	0.5	410	72	61SQ4732
HAQ1N190	190	4.4	0.5	399	75	61SQ4581
HAP1N155	155	4.3	1.3	408	72	61SP4467
HAP1N120	120	4.4	1.3	394	75	61SP4366
HAP1N95	95	4.5	1.3	377	79	61SP4290
HAP1N76	76	4.4	1.3	391	78	61SP4231
HAP1N60	60	4.3	1.3	411	75	61SP4183
HAP1N48	48	4.3	1.3	412	76	61SP4145
HAP1N37	37	4.7	1.3	345	91	61SP4113
HAC1N21.3	21.3	4.7	3.9	338	89	61SC5651
HAC1N13.5	13.5	4.9	3.9	326	95	61SC5409
HAC1N8.5	8.5	5.5	3.9	259	124	61SC5258
HAC1N5.3	5.3	6.8	3.9	166	192	61SC5162
HAC1N3.3	3.3	6.4	3.9	171	185	61SC5102
HAC1N2	2.0	8.1	3.9	119	294	61SC6640

TABLE 4 RECOMMENDED COLD LEAD CABLES FOR HAX MI SERIES HEATING CABLES

Nom. cross section [mm ²]	Reference Single Conductor Cable	Reference Dual Conductor Cable	Max. Current Design B* (single cond.)	Max. Current Design D. E* (dual cond.)	Nominal Diameter Single cond. (mm)	Diameter Dual cond. (mm)
1.0	–	AC2H1.0	–	18	–	7.3
2.5	AC1H2.5	AC2H2.5	34	28	5.3	8.7
6.0	AC1H6	AC2H6	57	46	6.4	14.0
16	AC1H16	AC2H16	102	98	9.0	14.7
25	AC1H25	AC2H25	133	128	10	17.1

All cold lead cables can be used for up to 600 Vac and use copper conductors with a temperature coefficient of 3.9×10^{-3} 1/K.

* For details on the different heating unit designs, refer to the chapter MI Heating Systems - MI Heating Cables in the Databook.

Delivery length of bulk cable on coil depends on type of resistance and is limited by max. coil length as indicated in the table on top. Factory terminated elements are limited by a max. weight of 50kg, however to ensure practical and safe on-site handling, it is strongly recommended to limit element lengths to 25 - 30kg. Not all resistances are standard items and as such may not be in stock. Contact Pentair to confirm lead time. Pentair requires the use of a 30 mA residual current device to provide maximum safety and protection from fire.

Where design results in higher leakage current, the preferred trip level for adjustable devices is 30 mA above any inherent capacitive leakage characteristic of the heater as specified by the trace heater supplier or alternatively, the next common available trip level for non adjustable devices, with a maximum of 300 mA. All safety aspects need to be proven.

TABLE 5 CHEMICAL RESISTANCE

Alloy	Maximum Cable Sheath Temp (°C)	Description	Nominal chemical composition, % (major elements)				High temperature resistance (+540°C)		Corrosion resistance									
INCOLOY Alloy 825 nickel-iron-chromium	550°C*	Excellent resistance to a wide variety of corrosives. Resists pitting and intergranular type corrosion, reducing acids and oxidizing chemicals	Nickel (+Cobalt)	Iron	Chromium	Other	Oxidation	Carburization	Sulfuric acid	Hydrochloric acid	Hydrofluoric acid	Phosphoric acid	Nitric acid	Organic acid	Alkalies	Salts	Seawater	Chloride cracking
			42.0	30.0	21.5	Mo 3.0 Cu 2.2	G-E	G-E	G-E	G-E	G-E	G-E	G-E	G-E	G-E	G-E	G-E	G-E

From Huntington Alloys Publication 78-348-2

Note: NR Not recommended, A acceptable, GE Good to excellent, X Check for specific data

* Temperature limitation based on construction of heating element.

Corrosion resistance data is dependent on temperature and concentration.



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